ThermalScout-Net | Current Project Review

COSC 2345 - Introduction to Artificial Intelligence | October 28, 2026 | Status: Synthetic proof of concept completed

29,635	8	0.9574 → 0.0220	3 of 3
Trainable parameters	Training epochs	Training loss	Sample objects detected

Project Objective

Develop and evaluate a custom computer-vision model that locates humans and animals in nighttime thermal drone imagery. The intended application is wildlife-conservation screening and anti-poaching support. The system will identify visible objects and draw bounding boxes; it will not determine whether a person is a poacher.

What Has Been Completed

• Created and ran thermal_scout_demo.py, a custom PyTorch convolutional neural network trained on generated thermal-style images.

• Trained the 29,635-parameter model from scratch on a CPU for eight epochs. Training loss dropped from 0.9574 to 0.0220 (about 97.7%).

• Produced and displayed predictions for a synthetic sample containing two humans and one animal; all three received the correct class and an enclosing box in the sample output.

• Saved demo weights to thermal_scout_demo_weights.pt. Previous BIRDSAI data preparation and a YOLO-based baseline script are available from earlier work.

Current Results and Limitations

The latest example displays approximately 97% prediction confidence for each of the three objects. These are confidence scores on synthetic examples, not real-world accuracy, precision, recall, or mAP. Decreasing training loss shows that the demo learned its synthetic training patterns, but it has not demonstrated generalization to unseen BIRDSAI footage. The current demo is a small proof-of-concept CNN, not yet the full proposed ThermalScout-Net detector with multi-scale feature fusion.

Planned Research Design

Use the BIRDSAI Conservation Drones dataset of labeled nighttime thermal videos; preserve a video-level separation between training, validation, and testing to avoid overlapping scenes. Compare (1) the full custom detector trained from random initialization, (2) the same design with a pretrained feature extractor, and (3) a fine-tuned YOLO11n baseline. Measure per-class precision and recall, mAP, false detections, inference speed, and training cost.

Immediate Next Steps

1. Confirm the actual BIRDSAI folder and label formats, then build a frame/annotation loader and visualize real labeled images.

2. Implement and debug the proposed real-image detection architecture; first test whether it can overfit a few labeled images.

3. Train all three comparable experiments with the same data split; report numerical metrics, failure cases, and trade-offs.

Current conclusion: The synthetic CNN demonstration and training pipeline work. The next milestone is evidence of detection performance on real, unseen thermal drone footage; no such performance claim has been established yet.